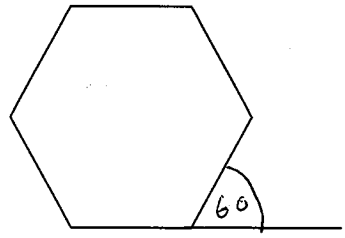
1 Work out the size of an exterior angle of a regular hexagon.



(Exterior angles always add to 360°)

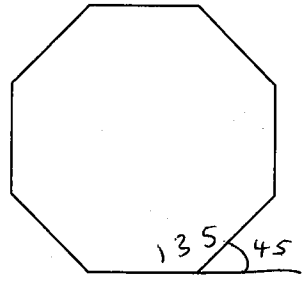
$$\frac{360}{6} = 60$$

.....60.....°

(Total for question 1 is 2 marks)

---

2 Work out the size of each interior angle in a regular octagon.



$$\frac{360}{8} = 45$$

$$180 - 45 = 135$$

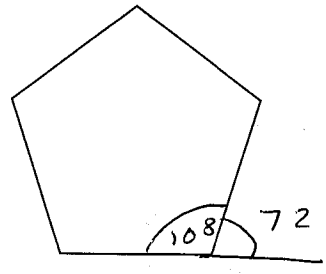
(Interior angle + Exterior angle = 180)

.....135.....°

(Total for question 2 is 2 marks)

---

3 Work out the size of each interior angle in a regular pentagon



$$\frac{360}{5} = 72$$

$$180 - 72 = 108$$

.....108.....°

(Total for question 3 is 2 marks)

4 The size of each exterior angle in a regular polygon is 20°.
Work out how many sides the polygon has.

$$\frac{360}{20} = \frac{36}{2} = 18$$

.....18.....

(Total for question 4 is 2 marks)

---

5 The size of each exterior angle in a regular polygon is 18°.
Work out how many sides the polygon has.

$$\frac{360}{18} = \frac{180}{9} = 20$$

.....20.....

(Total for question 5 is 2 marks)

---

6 The size of each interior angle in a regular polygon is 165°.
Work out how many sides the polygon has.

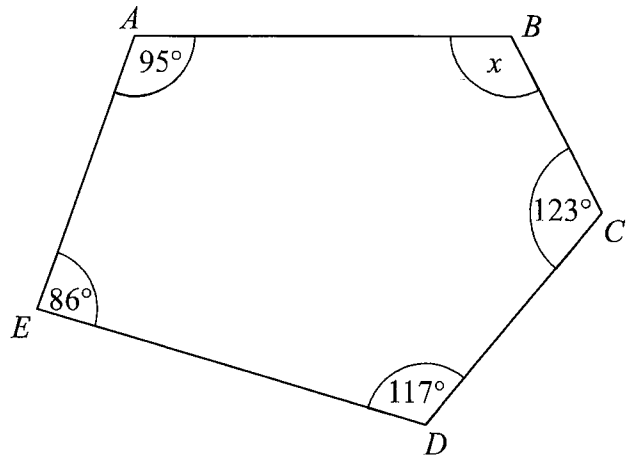
$180 - 165 = 15$ (Exterior angle = 15°)

$$\frac{360}{15} = \frac{120}{5} = 24$$

.....24.....

(Total for question 6 is 2 marks)

7



$$\begin{array}{r} 123 \\ 117 \\ 95 \\ +\ 86 \\ \hline 421 \end{array}$$

ABCDE is a pentagon.

Work out the size of angle *ABC*.

$540 - 421$
$= 119$

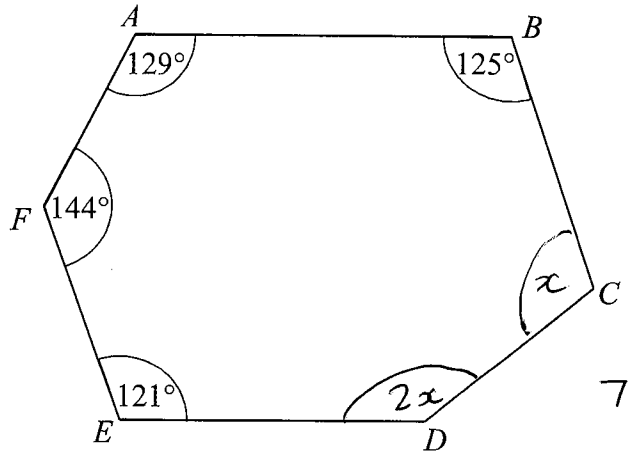
Angle sum $= (n-2) \times 180$
$= (5-2) \times 180$
$= 3 \times 180$
$= 540$

(Angles in a pentagon add to 540°)

119 °

(Total for question 7 is 2 marks)

8



$$\begin{array}{r} 129 \\ 125 \\ 144 \\ +\ 121 \\ \hline 519 \end{array}$$

$720 - 519 = 201$

$$\begin{array}{r} 67 \\ 3\overline{)201} \end{array}$$

ABCDEF is a hexagon.

Angle *CDE* = 2 × Angle *BCD*

Work out the size of angle *CDE*.

$x = 67$

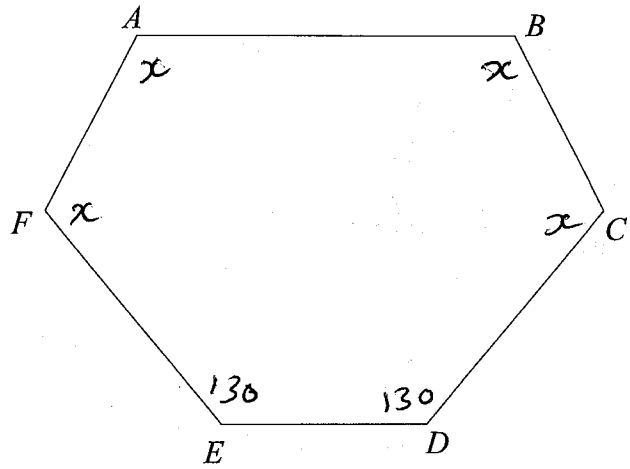
$2x = 2 \times 67$
$= 134$

Angles in a hexagon $= (6-2) \times 180$
$= 4 \times 180$
$= 720°$

134 °

(Total for question 8 is 3 marks)

9



ABCDEF is a hexagon.

Angle *BAF* = Angle *ABC* = Angle *AFE* = Angle *BCD*.
Angle *DEF* = Angle *CDE* = 130°

Work out the size of angle *BAF*.
You must show all your working.

Angles in a hexagon = $(6-2) \times 180$
$= 720°$

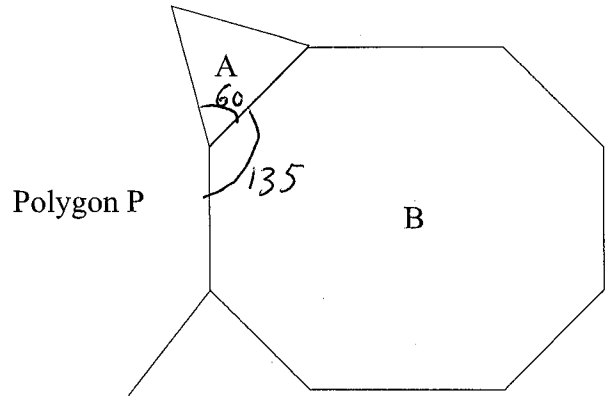
$720 - 130 - 130$
$720 - 260 = 460$

$\frac{460}{4} = \underline{115°}$

.....115.....°

(Total for question 9 is 3 marks)

10



Shape A is a regular triangle. Shape B is a regular octagon.

Another regular polygon, P, is shown on the diagram.

How many sides does polygon P have?

You must show your working.

Interior angle of A = 60°

Exterior angle of B = $\frac{360}{8}$ = 45°

Interior angle of B = 180 − 45

= 135°

Interior angle of P = 360 − 60 − 135

= 165°

Exterior angle of P = 180 − 165

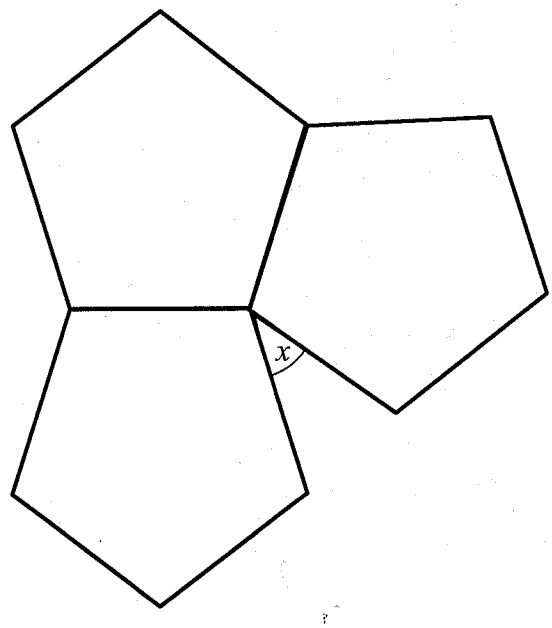
= 15°

$\frac{360}{15} = \frac{120}{5} = 24$

.....24.....

(Total for question 10 is 4 marks)

11



The diagram shows three regular pentagons meeting at a point.

Work out the size of the angle marked x.
You must show all your working.

Exterior angle of pentagon $= \frac{360}{5} = 72$

Interior angle of pentagon $= 180 - 72$
$= 108°$

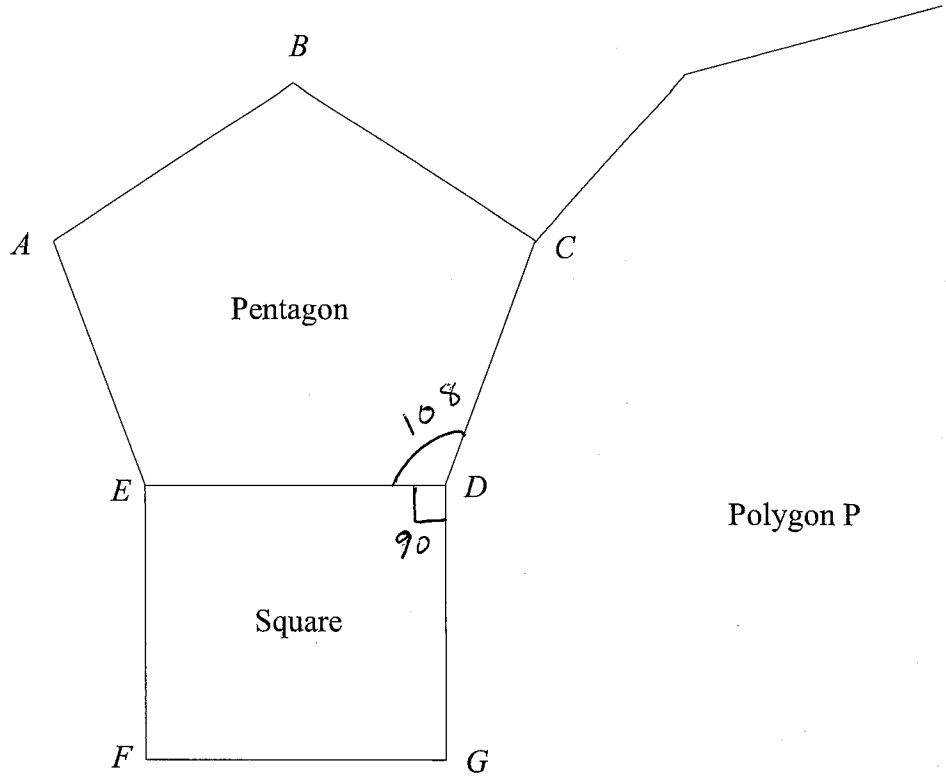
$3 \times 108 = 324$

$360 - 324 = \underline{\underline{36°}}$

.....36.....°

(Total for question 11 is 3 marks)

12



Polygon P

The diagram shows a regular pentagon, ABCDE, and a square, EDFG.

The lines CD and DG are both sides of another regular polgon, P.

How many sides does polygon P have?

You must show how you got your answer.

Exterior angle of pentagon $= \frac{360}{5} = 72$

Interior angle of pentagon ~~= 36~~

$= 180 - 72$

$= 108$

Interior angle of P $= 360 - 90 - 108$

$= 162°$

Exterior angle of P $= 180 - 162$

$= 18$

$\frac{360}{18} = 20$

20

(Total for question 12 is 4 marks)